A

Smart Irrigation System

Project Report

submitted

in partial fulfilment

for the award of the degree of

Bachelor of Technology

in department of Computer Science

with specialization in Computer Science

विद्या ददाति विनयम्

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[2025-2026]

ACKNOWLEDGEMENT

We would like to first of all express our thanks to Dr. Rajesh Bhardwaj, Group Director of LIET, for providing us such a great infrastructure and environment for our overall development.
We express sincere thanks to Principal Dr. Amit Kumar Malik for his kind cooperation and extensible support toward the completion of our seminar.
Words are inadequate in offering our thanks to Dr. Pratap Singh Patwal, HOD of CSE Department, for consistent encouragement and support for shaping our seminar in the presentable form. Also, for his support in providing technical requirement and fulfilling our various other requirement for making our seminar success.
We would like to thanks our project coordinator Mr. Atul Gaur, Asst. Prof. CSE Department for their support and guiding us to make seminar successful. Without their support and measurement, we would not able to accomplished our goal.
We also like to express our thanks to all supporting CSE faculty members who have been a constant source of encouragement for successful competition of the project.
Also, our warm thanks to Laxmi Devi Institute of Engineering and Technology, who provide us this opportunity to carry out this prestigious seminar and enhance our learning in various technical fields.

CANDIDATE'S DECLARATION

I hereby declare that the Industrial/Field Project, entitled "**Smart Irrigation System**" in partial fulfilment for the award of Degree of "**Bachelor of Technology**" in **Department of Computer Science & Engineering** with Specialization in Computer Science Engineering , and submitted to the Department of Computer Science & Engineering, **Laxmi Devi Institute of Engineering** & **Technology, Alwar, Bikaner Technical University** is a record of my own Learning and internship carried under the Guidance of Dr. Pratap Singh Patwal, HOD(CSE) : Laxmi Devi institute of Engineering & Technology, Alwar.

I have not submitted the matter presented in this field work report anywhere for the award of any other Degree.

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ABSTRACT

The Smart Irrigation System using Internet of Things (IoT) is designed to automate the irrigation process in agricultural fields by monitoring soil moisture and environmental conditions. Traditional irrigation methods often result in excessive water usage and require continuous human supervision. This project aims to reduce water wastage and improve efficiency through automation.

The system is built using an Arduino UNO microcontroller, soil moisture sensor, humidity sensor (DHT11), relay module, and a water pump. The soil moisture sensor continuously measures the water content in the soil, and when the moisture level falls below a predefined threshold, the system automatically activates the water pump through a relay. Once the soil reaches sufficient moisture level, the pump is turned off.

An LCD display (RG1602A) is used to show real-time sensor values, providing a user-friendly interface. The system is simple, cost-effective, and suitable for small-scale agricultural applications.

This project demonstrates how IoT-based automation can optimize water usage, reduce manual effort, and improve crop productivity.

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1. INTRODUCTION

1.1 Overview

Agriculture plays a vital role in the economy, and efficient water management is one of the major challenges faced by farmers. Traditional irrigation systems depend on manual operation, which often leads to over-irrigation or under-irrigation. This results in water wastage, reduced crop yield, and increased labour effort.

With the advancement of Internet of Things (IoT) technologies, it has become possible to automate irrigation systems using sensors and microcontrollers. Smart irrigation systems can monitor soil conditions in real-time and make decisions automatically without human intervention.

This project presents a Smart Irrigation System using Arduino UNO and sensors such as soil moisture sensor and DHT11 humidity sensor. The system continuously monitors the moisture level in the soil and activates a water pump when required. An LCD display is used to show real-time data, making the system user-friendly and efficient.

1.2 Motivation

The motivation behind developing the Smart Irrigation System arises from the increasing problem of water scarcity and inefficient irrigation methods.

- Farmers often irrigate fields manually without knowing actual soil moisture levels
- This leads to over-irrigation or under-irrigation
- Water resources are wasted unnecessarily

With the advancement in IoT and embedded systems, it is possible to develop low-cost automation systems that can improve irrigation efficiency. This project aims to utilize these technologies to create a smart and reliable irrigation solution.

1.3 Problem Statement

Traditional irrigation methods suffer from several limitations:

1. Excessive water usage due to lack of monitoring
2. Manual effort required to operate irrigation systems
3. Inconsistent watering affecting crop health
4. No real-time feedback on soil condition

There is a need for an automated system that can:

- Monitor soil moisture
- Control irrigation intelligently

- Reduce water wastage

1.4 Objective

The main objectives of the Smart Irrigation System are:

1. To design an automated irrigation system using Arduino
2. To monitor soil moisture using sensors
3. To control water pump using relay module
4. To display real-time data using LCD
5. To reduce water wastage and manual effort
6. To improve irrigation efficiency

1.5 Methodology

The Smart Irrigation System follows a systematic approach for monitoring and controlling irrigation using sensors and a microcontroller. The methodology of the system can be divided into the following steps:

1. Data Collection
The soil moisture sensor continuously measures the moisture level present in the soil. The DHT11 sensor is used to monitor environmental humidity.

2. Data Processing
The collected data from sensors is sent to the Arduino UNO microcontroller. The Arduino processes the input values and compares them with predefined threshold levels.

3. Decision Making
If the soil moisture level falls below a certain threshold (e.g., 30%), the system decides to turn ON the irrigation system. If the moisture level exceeds the upper limit (e.g., 70%), the system turns OFF the pump.

4. Actuation
The relay module acts as a switch to control the water pump. Based on the decision, the relay either activates or deactivates the pump.

5. Display Output
The LCD (RG1602A) displays real-time values of soil moisture and humidity, providing a user-friendly interface.

2. LITERATURE REVIEW & TECH STACK

2.1 Background Study and Existing Systems

Agriculture has always relied on irrigation systems to provide water to crops. Traditional irrigation methods such as manual watering and fixed-time irrigation systems are widely used but suffer from major drawbacks. These methods do not consider real-time soil conditions, which leads to inefficient water usage.

Earlier systems used timer-based irrigation, where water is supplied at fixed intervals regardless of soil moisture. This often results in over-irrigation or under-irrigation. With the advancement of technology, sensor-based irrigation systems have been developed to improve efficiency.
Modern smart irrigation systems use soil moisture sensors to monitor soil conditions and automatically control water supply. Some advanced systems also integrate IoT platforms to enable remote monitoring and control. However, many of these systems are expensive and require complex infrastructure.
The proposed system focuses on developing a low-cost, efficient, and easy-to-implement solution using Arduino and basic sensors.

2.2 Tech Stack Deep Dive

This project uses a combination of hardware components and embedded system technologies to implement an efficient and automated irrigation system. Each component plays a crucial role in sensing, processing, and controlling the irrigation process.

1. **Arduino UNO (Microcontroller)**
Arduino UNO is the central controlling unit of the system. It is based on the ATmega328P microcontroller and is responsible for processing input data from sensors and controlling output devices such as relay and LCD.
The Arduino continuously reads analog data from the soil moisture sensor and digital data from the DHT11 sensor. Based on this data, it makes decisions according to predefined conditions and controls the irrigation system accordingly.

Key Features:

- 14 digital input/output pins and 6 analog input pins
- Operates at 5V supply
- Clock speed of 16 MHz
- Easy programming using Arduino IDE

Advantages:

- Low cost and widely available
- Easy to interface with sensors and modules
- Large community support and documentation

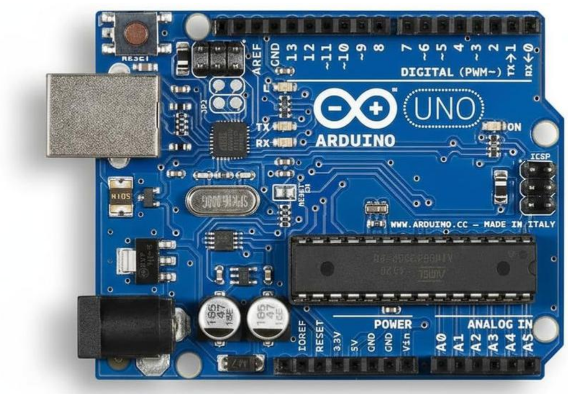

Fig 1: Arduino UNO Microcontroller

2. Soil Moisture Sensor

The soil moisture sensor is one of the most important components in the system. It is used to measure the volumetric water content in the soil.

The sensor consists of two conductive probes that are inserted into the soil. When voltage is applied, the resistance between the probes changes depending on the moisture level.

Working Principle:

- Wet soil → better conductivity → lower resistance
- Dry soil → poor conductivity → higher resistance

The sensor outputs an analog signal, which is read by the Arduino through the analog pin (A0). This value is then converted into a percentage of moisture.

Applications:

- Agriculture monitoring
- Smart irrigation systems
- Soil condition analysis

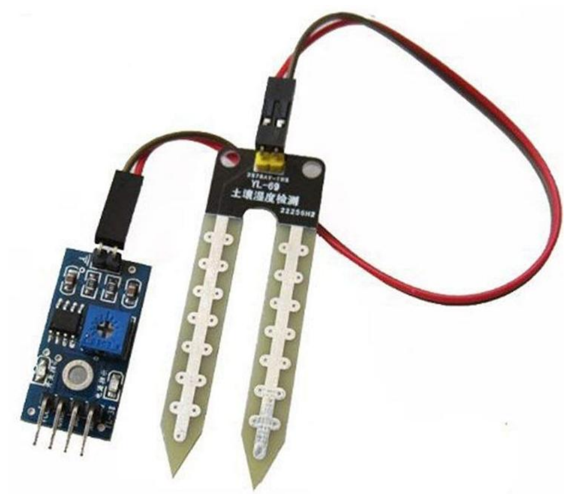

Fig 2: Soil Moisture Sensor

3. DHT11 Humidity Sensor

The DHT11 sensor is used to measure environmental humidity and temperature. It provides digital output, making it easy to interface with Arduino.
The sensor contains a humidity sensing component and a thermistor. It converts the measured data into a calibrated digital signal.

Working:

- Measures humidity in percentage (%)
- Sends data to Arduino via single data pin

Features:

- Low power consumption
- Simple interface
- Reliable for basic environmental monitoring

Role in Project:

The DHT11 sensor helps in understanding environmental conditions, which can indirectly affect irrigation decisions.

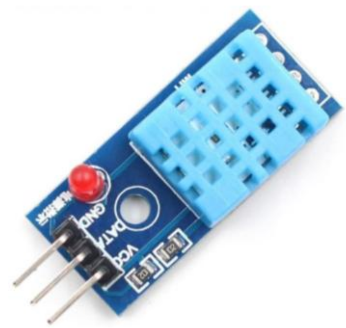

Fig 3: DHT11 Humidity Sensor

4. Relay Module

The relay module acts as an electrically operated switch. It allows the Arduino to control high-power devices like the water pump.
Since Arduino operates at low voltage (5V), it cannot directly drive high-power components. The relay provides isolation and safe switching.

Working Principle:

- Arduino sends signal to relay input
- Relay coil gets energized
- Internal switch closes → pump turns ON
- When signal is removed → switch opens → pump OFF

Types:

- Normally Open (NO)
- Normally Closed (NC)

In this project, the **Normally Open (NO)** configuration is used.

Fig 4: Relay Module

5. **Water Pump**

The water pump is the output device responsible for supplying water to the soil. It is controlled through the relay module.
When the relay is activated, the pump starts drawing water and supplies it to the irrigation area. Once sufficient moisture is achieved, the pump is turned OFF.

Features:

- Small size and portable
- Low power consumption
- Suitable for small irrigation setups

Applications:

- Garden watering systems
- Automatic irrigation
- Small-scale agriculture

Fig 5: Water Pump

6. **LCD Display (RG1602A)**

The LCD display is used to show real-time information such as soil moisture level and humidity.
It is a 16x2 display, meaning it can display 16 characters per line and has 2 lines.

Working:

- Connected using I2C interface (SDA, SCL)
- Receives processed data from Arduino
- Displays sensor values continuously

Advantages:

- Easy to read output
- Improves user interaction
- Helps in debugging and monitoring

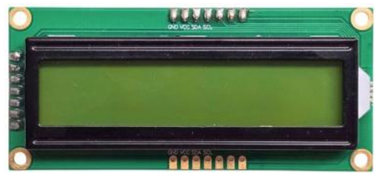

Fig 6: LCD Display (RG1602A)

7. Power Supply (Battery)

The power supply is an essential component of the smart irrigation system. It provides the required electrical energy to operate the Arduino, sensors, relay module, and water pump.

In this project, a battery is used as the power source to ensure portability and ease of use. The Arduino UNO operates at 5V, while the relay module and sensors also require low voltage power. The battery ensures continuous operation of the system without dependency on external power supply.

For stable operation, voltage regulation is important to avoid damage to components. The battery can be connected through a voltage regulator or directly through the Arduino power input depending on the voltage rating.

The power supply provides the necessary electrical energy to run all components of the system.

In this project, a battery is used to make the system portable and independent of external power sources. The Arduino and sensors require a stable 5V supply for proper operation.

Working:

- Battery supplies voltage to Arduino
- Arduino distributes power to sensors and modules
- Relay controls power to pump separately

Important Consideration:
Voltage regulation is important to prevent damage to components and ensure stable performance.

Fig 7: 9V Battery

3. SYSTEM ANALYSIS & DESIGN

3.1 Requirements Specification

1 Hardware Requirements

The following hardware components are required for the Smart Irrigation System:

- Arduino UNO
- Soil Moisture Sensor
- DHT11 Humidity Sensor
- Relay Module
- Water Pump
- LCD Display (16x2 – RG1602A)
- Battery / Power Supply

Each component plays a specific role in sensing, processing, and controlling the irrigation system.

2 Software Requirements

The software tools required for implementation are:

- Arduino IDE (for programming and uploading code)
- Embedded C programming language
- Sensor libraries such as:
    - ❖ DHT Library
    - ❖ LiquidCrystal / I2C Library

The software is responsible for reading sensor data, processing it, and controlling the output devices.

3 Functional Requirements

Functional requirements define what the system should do:

- Continuously monitor soil moisture level
- Measure environmental humidity using DHT11
- Automatically control water pump based on moisture level
- Display real-time data on LCD
- Operate without human intervention

3.2 System Architecture

The system architecture is designed as a closed-loop automated control system. It consists of three main parts:

1. Input Layer
The input layer includes sensors such as the soil moisture sensor and DHT11 sensor. These sensors continuously collect data from the environment.

- Soil moisture sensor detects water content in soil
- DHT11 measures humidity and temperature

2. Processing Layer
The Arduino UNO acts as the processing unit. It receives data from sensors and processes it using predefined logic.

- Converts analog signal to digital values
- Compares values with threshold limits
- Makes decisions based on conditions

3. Output Layer
The output layer consists of relay module, water pump, and LCD display.

- Relay controls the water pump
- Pump supplies water to soil
- LCD displays real-time data

4. Power Layer
The power layer ensures proper functioning of all components.

- Battery supplies power
- Voltage is regulated for stability
- Ensures continuous system operation

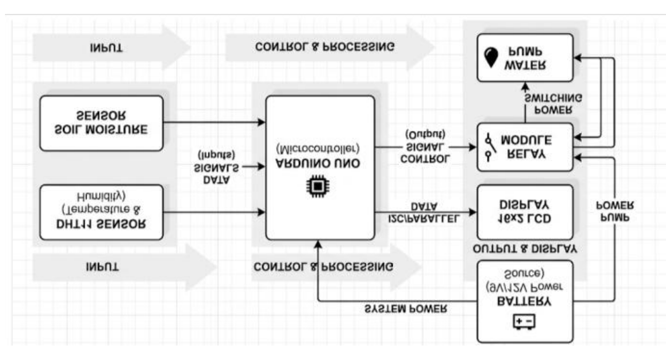

Fig 8: Block Diagram of Smart Irrigation System

3.3 Data Flow/Work Flow

The system starts by initializing all components. The Arduino continuously reads data from the soil moisture sensor and humidity sensor.

If the soil moisture value falls below the lower threshold (30%), the system activates the water pump through the relay module. If the moisture level exceeds the upper threshold (70%), the pump is turned off.

The process runs continuously in a loop, ensuring real-time monitoring and control. The LCD display shows current values, helping users understand system operation.

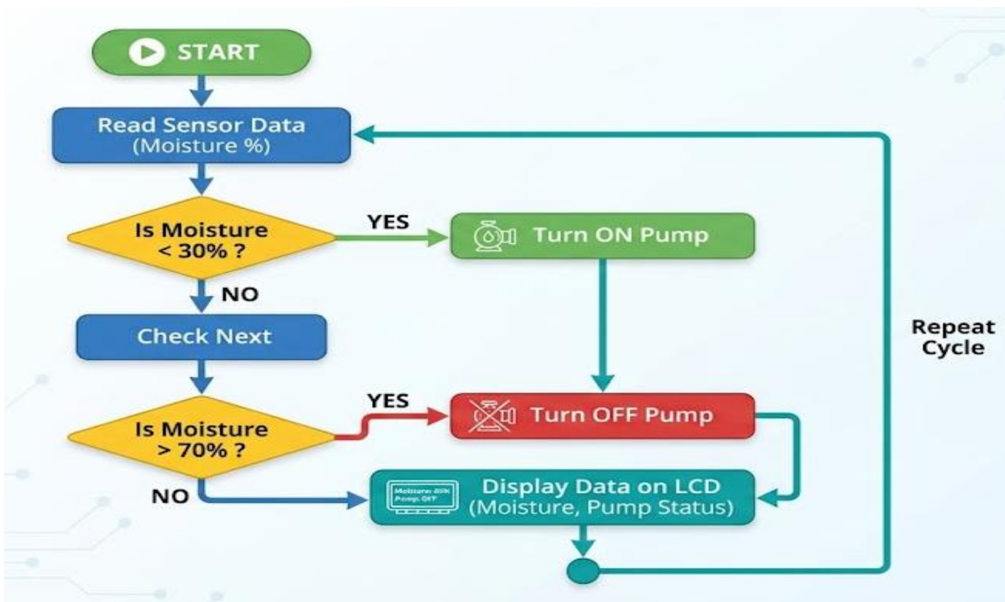

Fig 9: Flowchart of Smart Irrigation System

The system follows a continuous loop-based execution:

1. The system starts and initializes all components
2. Arduino reads sensor values
3. Moisture level is compared with threshold values
4. If soil is dry → pump is activated
5. If soil is wet → pump is turned off
6. Data is displayed on LCD
7. Process repeats continuously

This ensures real-time monitoring and automatic irrigation.

3.4 System Constraints

The system has certain constraints that affect its performance:

- Limited sensor accuracy
- Dependence on battery power
- Suitable only for small-scale implementation
- No remote connectivity

4. IMPLEMENTATION

4.1 Implementation Environment Setup

The Smart Irrigation System is implemented using Arduino UNO as the central microcontroller. The programming is done using Arduino IDE with Embedded C language. All hardware components such as sensors, relay module, LCD display, and water pump are connected according to the circuit design. The system operates on a regulated 5V power supply to ensure stable and reliable performance.
The required libraries such as DHT library and LiquidCrystal (I2C) library are included in the program to interface sensors and display modules.

4.2 Circuit Connections

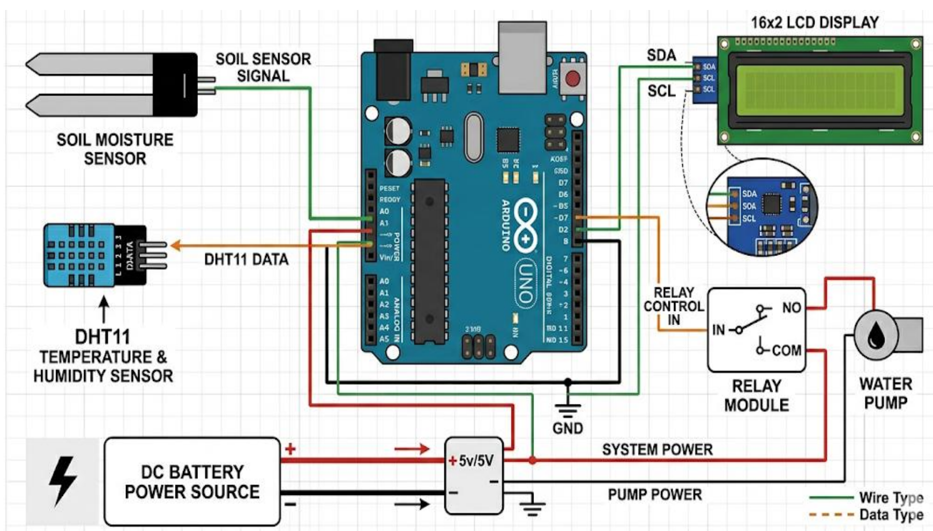

Fig 10: Circuit Diagram of Smart Irrigation System

Detailed Pin Configuration

Component	Arduino Pin
Soil Moisture Sensor	A0
DHT11 Sensor	D2
Relay Module	D7
LCD (I2C) SDA	A4
LCD (I2C) SCL	A5
VCC / GND	Power Supply

Explanation

The soil moisture sensor is connected to the analog pin A0 of Arduino to read moisture values. The DHT11 sensor is connected to digital pin D2 for humidity measurement.

The relay module is connected to digital pin D7, which controls the switching of the water pump. The LCD display is connected using I2C communication for displaying real-time data. The water pump is connected to the relay module, which acts as a switch. A battery is used to power the system.

4.3 System Operation Logic

The system operates based on threshold-based decision logic.

- Lower Threshold: 30% (Dry condition)
- Upper Threshold: 70% (Wet condition)

Working Logic:

- If moisture < 30% → Pump ON
- If moisture > 70% → Pump OFF
- Else → No change

This logic ensures that water is supplied only when required, preventing wastage.

4.4 Algorithm

Algorithm: Smart Irrigation System

1. Start the system
2. Initialize Arduino, sensors, LCD, and relay module
3. Read soil moisture value from sensor
4. Read humidity value from DHT11 sensor
5. Display values on LCD
6. If (moisture < 30%)
   Turn ON water pump
7. Else if (moisture > 70%)
   Turn OFF water pump
8. Repeat steps 3 to 7 continuously
9. Stop

4.5 Pseudo Code

```
START

Initialize sensors and LCD
Set relay pin as OUTPUT

LOOP:
Read soil_moisture from A0
Read humidity from DHT11

Display values on LCD

IF soil_moisture < 30 THEN
Turn ON relay (pump ON)
ELSE IF soil_moisture > 70 THEN
Turn OFF relay (pump OFF)
END IF

REPEAT LOOP
```

4.6 Working Principle

The system works on the principle of automatic control using sensor feedback.
The soil moisture sensor detects the water content in the soil and sends analog signals to the Arduino. The Arduino processes this data and compares it with predefined threshold values. When the soil is dry, the Arduino activates the relay module, which turns ON the water pump. When the soil becomes sufficiently wet, the Arduino deactivates the relay, turning OFF the pump.
The LCD display continuously shows real-time sensor values, allowing users to monitor system performance.

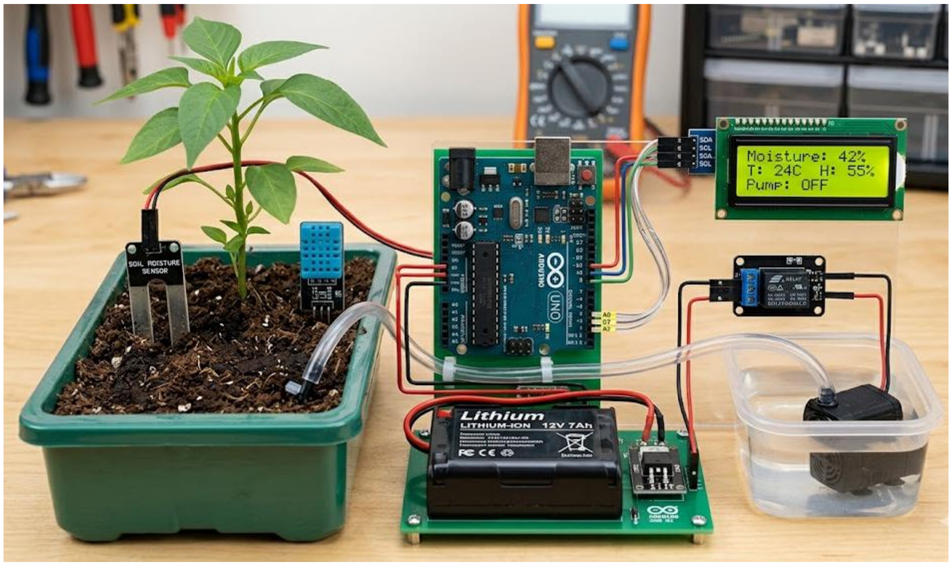

Fig 11: Actual Smart Irrigation System Setup

4.7 Advantages of Implementation

The Smart Irrigation System offers several advantages over traditional irrigation methods:

1. Water Conservation
The system ensures that water is supplied only when required, reducing unnecessary wastage. This helps in efficient utilization of water resources.

2. Automation
The system operates automatically without human intervention, reducing manual effort and increasing convenience.

3. Cost-Effective
The components used in the system are low-cost and easily available, making it affordable for small-scale farmers.

4. Real-Time Monitoring
The LCD display provides real-time information about soil moisture and humidity, allowing users to monitor system performance.

5. Energy Efficiency
The system consumes low power and can be operated using battery or solar power.

6. Improved Crop Health

Proper irrigation helps maintain soil moisture at optimal levels, improving plant growth and crop yield.

7. Easy Installation and Maintenance

The system is simple to set up and does not require complex infrastructure.

4.8 Comparison between Traditional and Smart Irrigation System

Feature	Traditional System	Smart Irrigation System
Water Usage	High	Optimized
Control	Manual	Automatic
Efficiency	Low	High
Monitoring	Not available	Real-time
Cost	Low (initial)	Moderate
Accuracy	Low	High

5. TESTING AND RESULTS

5.1 Testing Strategy

The Smart Irrigation System was tested to verify its functionality, accuracy, and reliability under different environmental conditions. The testing process ensures that all components such as sensors, microcontroller, relay module, and water pump are working correctly.
The following testing approaches were used:

1. Unit Testing
Each component was tested individually:

- Soil moisture sensor tested for different soil conditions
- DHT11 sensor tested for humidity readings
- Relay module tested for switching operation

2. Integration Testing
All components were connected and tested together to ensure proper communication between sensors, Arduino, and output devices.

3. Real-Time Testing
The system was tested in actual soil conditions to observe real-world performance and behaviour.

5.2 Performance Parameters

The performance of the system is evaluated based on the following parameters:

- **Accuracy**: Correct detection of soil moisture levels
- **Response Time**: Time taken to activate or deactivate pump
- **Reliability**: Consistent performance over repeated cycles
- **Power Efficiency**: Low power consumption during operation

5.3 Test Cases and Observations

Test Case	Condition	Expected Output	Actual Output	Status
TC01	Dry Soil (Moisture < 30%)	Pump ON	Pump ON	Pass
TC02	Wet Soil (Moisture > 70%)	Pump OFF	Pump OFF	Pass
TC03	Normal Soil (30–70%)	No change	Stable	Pass

TC04	Sensor Disconnected	No reading	Error/No output	Pass
TC05	System Power ON	LCD displays values	Display working	Pass

5.4 Performance Evaluation

- The system showed high accuracy in detecting soil moisture levels
- Fast response time ensured timely irrigation
- Reliable operation during repeated testing cycles
- Low power consumption made it suitable for battery operation

5.5 Graphical Analysis

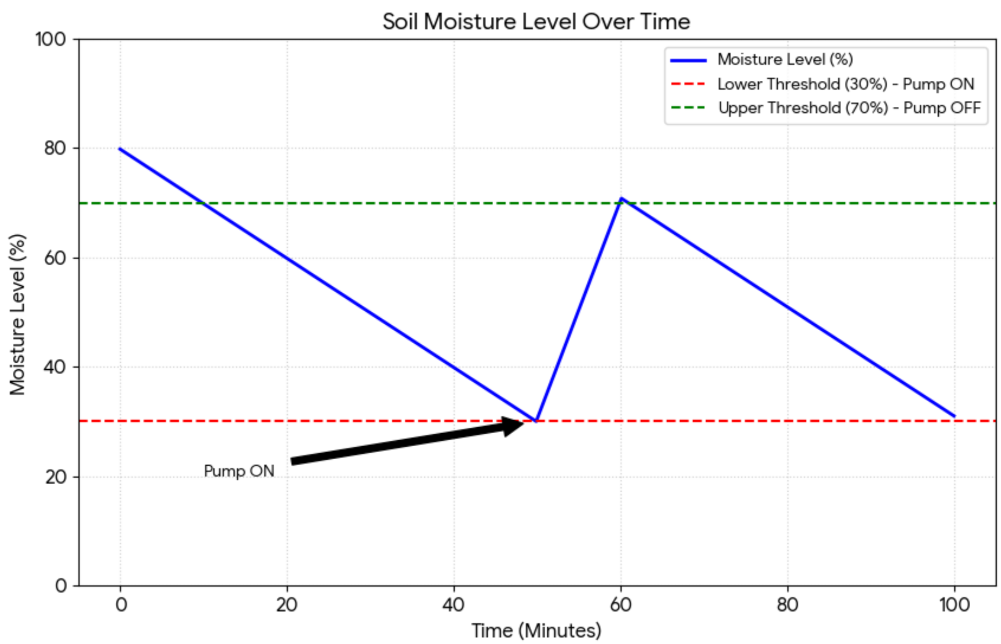

Fig 12: Moisture Level Variation Over Time

5.6 Results and Discussion

The Smart Irrigation System performed successfully under all testing conditions. The soil moisture sensor accurately detected changes in moisture levels and triggered the relay module accordingly.
The response time of the system was quick, and the pump was activated immediately when the soil became dry. Similarly, the system turned OFF the pump when sufficient moisture was achieved.
The LCD display provided real-time data, improving user interaction and monitoring. The system operated reliably without any major errors during continuous testing.

6. FUTURE SCOPE & LIMITATIONS

6.1 Introduction

The Smart Irrigation System provides an efficient solution for automated watering using sensors and a microcontroller. While the system performs well for small-scale applications, there are certain limitations that affect its performance. At the same time, the system has strong potential for further improvements and real-world applications.

1.2 Limitations

- **Limited Range of Sensors**
  The system uses basic sensors, which may not provide highly accurate readings in all environmental conditions.
- **No Remote Monitoring**
  The system does not support mobile or web-based monitoring, limiting user control to physical setup only.
- **Manual Power Dependency**
  The system depends on battery power, which requires regular replacement or charging.
- **Small-Scale Implementation**
  The system is suitable for small areas and may not be efficient for large agricultural fields.
- **No Weather Prediction Integration**
  The system does not consider weather conditions like rainfall, which could affect irrigation decisions.

1.3 Future Scope

1. **IoT Integration**
   The system can be integrated with Internet of Things (IoT) platforms such as Blynk or ThingSpeak to enable remote monitoring and control.

   This would allow users to:
   - Monitor soil moisture levels in real-time
   - Control irrigation from a smartphone
   - Receive alerts when moisture is low

2. **Solar Power System**
   The system can be enhanced by integrating solar panels as a power source. This will make the system energy-efficient and suitable for rural areas where electricity supply is limited.

Advantages:

- Renewable energy usage
- Reduced operational cost
- Environment-friendly solution

3. **Mobile Application Control**
A dedicated mobile application can be developed to provide a better user interface. Through the app, users can:
    - View live sensor data
    - Control pump manually if required
    - Set custom moisture thresholds
    - This improves usability and control over the system.

4. **Multi-Sensor Integration**
The system can be expanded by adding more sensors such as:
    - Temperature sensor
    - Rain sensor
    - Water level sensor
    - These sensors will provide additional data, enabling more accurate irrigation decisions.
    - Example:
    If rain is detected → irrigation can be stopped automatically

5. **Artificial Intelligence Based Irrigation**
The system can be upgraded using Artificial Intelligence (AI) and Machine Learning techniques. AI models can analyze historical data and predict irrigation requirements.
Benefits:

- Smart decision making
- Predictive irrigation
- Improved crop yield

6. **Large Scale Deployment**
The system can be extended for large farms by using multiple sensors and distributed nodes. Each section of the field can have its own sensor and irrigation control system.
    - This will enable:
    - Zone-wise irrigation
    - Efficient water distribution

- Better farm management

7. Cloud Data Storage and Analytics

The system can be connected to cloud platforms to store data for long-term analysis. Farmers can analyze:

- Moisture trends
- Water usage patterns
- Crop performance

This helps in improving future irrigation strategies.

8. Automatic Fertilizer System

The system can be upgraded to include fertilizer dispensing along with irrigation. This is known as **fertigation**.

Advantages:

- Saves time and labor
- Ensures proper nutrient supply
- Improves crop quality

9. Wireless Communication System

Wireless technologies like Wi-Fi, Bluetooth, or GSM can be used to make the system completely wireless. This will eliminate the need for physical connections and improve flexibility.

7. CONCLUSION

The Smart Irrigation System developed in this project successfully demonstrates the application of embedded systems and sensor-based automation in agriculture. The system effectively monitors soil moisture and environmental conditions using sensors and controls the irrigation process automatically.

By using Arduino UNO, soil moisture sensor, DHT11 sensor, relay module, and water pump, the system ensures efficient water usage and reduces the need for manual intervention. The automatic switching mechanism helps prevent over-irrigation and under-irrigation, improving overall crop health.

The use of an LCD display provides real-time monitoring, making the system user-friendly and easy to operate. The project is cost-effective, simple to implement, and suitable for small-scale agricultural applications.

Overall, the Smart Irrigation System proves to be a reliable and efficient solution for modern irrigation challenges. It also lays the foundation for future enhancements using IoT and advanced automation technologies.

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